

Individual Reflection: Lab 4 – Cloud ML Platform Analysis

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Group Number: 04

My Tasks: Strengths and Limitations Summary; Conclusion

1. What I did in the group

In our group of four, we split the work based on the different parts of a machine learning project. My specific job was to:

- Summarize the Pros and Cons: I looked at all the research we did on AWS, Google, and Azure, and then I made a table to show what each platform is good at and where they struggle.
- Write the Conclusion: I wrote the final part of the report and created a "Recommendation Matrix". This helps people quickly see which platform to pick depending on their situation, like if they are a small startup or a big healthcare company.

2. The challenges I faced

The biggest challenge for me was **trying to simplify all the technical details**.

- Finding the balance: Some platforms, like AWS SageMaker, can do almost everything, but they are also really hard to learn. I had to figure out how to explain that it's "powerful but complicated" without making the summary too long.
- Comparing different features: Each cloud provider has its own "special skill." For example, Azure is amazing for following government rules (like HIPAA), while Google is great because it connects so well with BigQuery. It took a few tries to explain these differences clearly so anyone could understand them.

3. Why ML Engineering is important

After doing this lab, I think ML Engineering is super important because:

- It keeps things running: In our report, we saw that platforms have tools to monitor the model after it's launched. Without ML Engineering, a model might stop working correctly over time as data changes, and nobody would notice.

- It makes things scale: For a case like healthcare fraud, you might have thousands of claims coming in at once. ML Engineering makes sure the system is strong enough to handle all that work without crashing.
- It keeps things fair: In healthcare, it's not enough to just be "right"—you have to be fair and follow the law. ML Engineering includes tools like Azure's Responsible AI to make sure the model isn't biased against certain people.

4. Conclusion

This project taught me that there isn't one "best" platform for everyone. The right choice depends on how much experience your team has and what kind of data you are using. Doing this summary helped me understand that being an ML engineer is as much about making smart choices as it is about writing code.